

Multibranch Machine Learning-Assisted Optimization and Its Application to Antenna Design

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Abstract—To satisfy certain requirements for an antenna design, many full-wave electromagnetic (EM) simulations are required, and they involve a considerable computational burden. A multibranch machine learning-assisted optimization (MB-MLAO) method is proposed to dramatically reduce the computational complexity in this task. Then, this method is applied to antenna design and worst case performance (WCP) searching under a practical manufacturing tolerance. In the conventional Gaussian process regression (GPR)-based MLAO method, a lower confidence bound (LCB) prescreening strategy with an empirical LCB constant is used to weigh the predicted value and predicted uncertainty. Using a variable-fidelity machine learning method, an adaptive LCB variable, and a retraining and repredicting method, the proposed MB-MLAO method can strike a delicate balance between exploitation and exploration in searching. Moreover, variable-fidelity data from full-wave EM simulations are used in the multifidelity GPR machine learning method to further reduce the computational burden. Finally, two test functions and three types of antennas are selected as examples to illustrate the superiority of the proposed MB-MLAO method.

Index Terms—Antennas, machine learning, optimization methods, worst case performance (WCP) searching.

I. INTRODUCTION

WITH the increasing performance specifications of modern antenna systems, especially in the millimeter-wave (mmWave) frequency band, considering the relationships of multiple parameters in optimization, the sensitivity analysis (SA) and robust design play an indispensable role in antenna and antenna array design [1]–[3]. Due to the

increasing complexities of antenna topology and antenna usage environments, full-wave electromagnetic (EM) simulators are generally used to provide reliable simulation results. However, it is difficult to satisfy the design targets in a short time through parameter scanning and manual adjustment using a full-wave EM simulator. Moreover, it is almost impossible to obtain analytical solutions for practical antenna optimization problems; therefore, metaheuristic algorithms, such as genetic algorithm (GAs) [4], differential evolution algorithms (DEAs) [5], and particle swarm optimization (PSO) [2], [6], have been utilized to solve such problems. Unfortunately, combining a metaheuristic algorithm with a full-wave EM simulator results in another problem: hundreds or thousands of function calls are made before converging to a good result [7]. Each function call requires a time-consuming full-wave simulation, which results in an unbearable computational burden.

An efficient method to reduce the calculation time is training a low-complexity surrogate model using machine learning to replace a very time-consuming EM model. Various machine learning-assisted optimization (MLAO) methods have been studied and applied to the field of EM device design and have achieved excellent results, such as artificial neural networks (ANNs) [8], K-nearest neighbor (KNN) algorithms [9], and support vector machines (SVMs) [10]. To further alleviate the computational budget, the variable-fidelity-based MLAO method was introduced [11]–[16]. The low-fidelity (LF) model can be an equivalent-circuit model or a coarse-discretization EM simulation model, and the high-fidelity (HF) model can be a high-simulation-accuracy EM model. Learning the relationship between data of different fidelities can achieve a higher precision surrogate model using a small amount of HF data.

Based on the online updating surrogate model mechanism in [17], Gaussian process machine learning (GPML) combined with DEA to assist optimization was applied [18]. Using the predicted value and predicted uncertainty provided by the GPML, a lower confidence bound (LCB) prescreening strategy was introduced to balance exploitation and exploration by controlling the LCB constant. The setting of the LCB constant was often based on the designer's experience [19]. The LCB constant was set to 1 in [13], while it was suggested to be equal to 2 in [19]. However, the antenna optimization problem varies case by case, and the setting of the LCB constant strongly impacts the performance and robustness of

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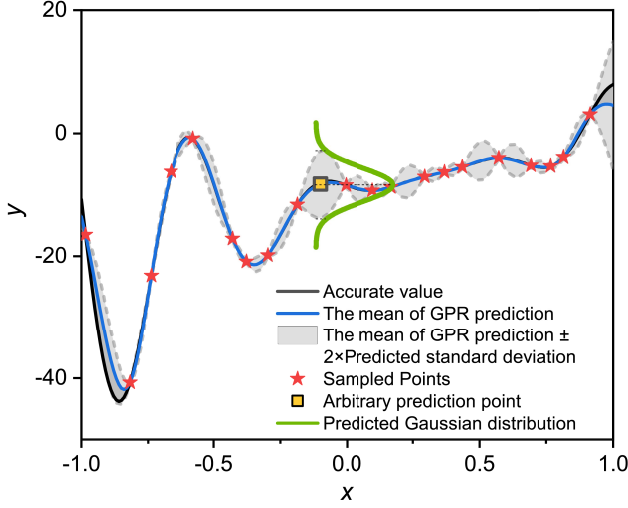


Fig. 1. Illustrated example of predicted values and predicted uncertainties using GPR.

the algorithm. A self-adaptive LCB (ALCB) combined with DEAs is proposed [20]. In this method, each individual of the population in DEA must evaluate the LCB constant. Some benchmark problems are experimented, and ALCB is efficient compared to other prescreening methods.

In this study, a multibranch machine learning-assisted optimization (MB-MLAO) method is proposed based on the application of adaptive LCB variables and a retraining and repredicting method. In the global optimization algorithm, the fitness function under different LCB variables is used to search through different branches to enhance the robustness of the MLAO algorithm. The data of the LF EM model are simulated to reduce the computational expenditure, and the multifidelity Gaussian process regression (GPR) model is retrained to improve prediction accuracy. Furthermore, the proposed method adaptively verifies one data point by the HF EM model and adds it to the sample set. The main contributions of this work are given as follows.

- 1) The MB-MLAO method balances between convergence speed and computational complexity and provides excellent adaptation regarding different characteristics and complexity antenna design problems.
- 2) Based on the surrogate model, the MB-MLAO method is applied using the LCB variable formed by multiple LCB constants to replace a fixed LCB constant to achieve a tradeoff between exploration and exploitation; then, multiple converged values and their corresponding input parameters are obtained.
- 3) The LF EM model is calculated to reduce the computational burden, and the surrogate model is retrained. The repredicted values are likely more accurate because there are corresponding LF data. Based on the repredicted values, one input vector is adaptively selected for the HF EM model to verify. The LCB constant corresponding to each verification value is uncertain; in each iteration, only one HF EM model is simulated.

The structure of this article is organized as follows. Section II briefly explains the mechanism of GPR, LCB

prescreening strategy, and multifidelity GPR and introduces the MB-MLAO algorithm in detail. Two nonlinear test functions and three practical antennas are used to verify the validation and adaptation of the proposed MB-MLAO in Sections III and IV, respectively. Finally, Section V concludes this article.

Notations: Upper (lower) boldface letters are used to denote matrices (vectors). Superscript $(\cdot)^T$ denotes the transpose. $\mathcal{N}(\mu, \sigma^2)$ denotes a Gaussian distribution with a mean μ and variance σ^2 .

II. MB-MLAO

In this section, GPR, the LCB prescreening strategy, and multifidelity GPR are described first. Next, the details of MB-MLAO method are presented.

A. Gaussian Process Regression

GPR is a machine learning method based on a Gaussian random process, a kernel trick, and Bayesian inference theory [21]. It has good adaptability for small samples, nonlinearities, and complex problems. GPR is used to obtain a low-cost surrogate model to replace function computation with high complexity. As the GPR example shown in Fig. 1, the predicted value and predicted uncertainty follow a Gaussian distribution. A brief introduction to GPR is presented as follows. More details of GPR are given in [21]–[23]. The training set is $\mathbb{D} = \{\mathbf{X}_{N \times K}, \mathbf{y}_{N \times 1}\}$, $\mathbf{X}_{N \times K} = \{\mathbf{x}^i | i = 1, \dots, N\}$, $\mathbf{x}^i = (x_1^i, x_2^i, \dots, x_K^i) \in \mathbb{R}^K$, and $\mathbf{y}_{N \times 1} = \{y^i | i = 1, \dots, N\}$, $y^i \in \mathbb{R}$, where N is the number of training samples and K is the dimension of the input vector. The GPR model is treated as

$$y^i = z(\mathbf{x}^i) + \epsilon^i = \boldsymbol{\phi}(\mathbf{x}^i)^T \boldsymbol{\beta} + \epsilon^i \quad (1)$$

where $z(\mathbf{x}^i)$ is a regression term, ϵ^i is an independent error term that follows the Gaussian distribution, $\boldsymbol{\phi}(\mathbf{x}^i)$ is a set of basic functions, and $\boldsymbol{\beta}$ is a set of regression coefficients. Assume that the following conditions hold.

- 1) $\boldsymbol{\phi}(\mathbf{x}^i) = \mathbf{1}$, $\boldsymbol{\beta} = \mu$, and $y^i = \mu + \epsilon(\mathbf{x}^i)$, $\epsilon(\mathbf{x}^i) \sim \mathcal{N}(0, \sigma^2)$.
- 2) The correlation of any two points $\epsilon(\mathbf{x}^i)$ and $\epsilon(\mathbf{x}^j)$ depends on the distance between the corresponding points, which is described as

$$\text{corr}(\epsilon(\mathbf{x}^i), \epsilon(\mathbf{x}^j)) = \exp\left(-\sum_{q=1}^K \theta_q |x_q^i - x_q^j|^{p_q}\right) \quad (2)$$

where μ , σ^2 , $\boldsymbol{\theta} = \{\theta_q | \theta_q > 0, q = 1, \dots, K\}$ and $\mathbf{p} = \{p_q | 1 \leq p_q \leq 2, q = 1, \dots, K\}$ are hyperparameters that must be estimated.

The likelihood function is given by

$$\mathcal{L}(\mu, \sigma^2, \boldsymbol{\theta}, \mathbf{p}) = \frac{\exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{1}\mu)^T \mathbf{C}^{-1}(\mathbf{y} - \mathbf{1}\mu)\right)}{(2\pi\sigma^2)^{\frac{N}{2}} |\det(\mathbf{C})|^{\frac{1}{2}}} \quad (3)$$

where $\mathbf{1}$ denotes an N -dimensional column vector of all ones and $\mathbf{C}$ is a correlation matrix of size $N \times N$. Two

hyperparameters μ and σ are estimated by using the maximum likelihood criterion for (3)

$$\hat{\mu} = \frac{\mathbf{1}^T \mathbf{C}^{-1} \mathbf{y}}{\mathbf{1}^T \mathbf{C}^{-1} \mathbf{1}} \quad (4)$$

$$\hat{\sigma}^2 = \frac{(\mathbf{y} - \mathbf{1}\hat{\mu})^T \mathbf{C}^{-1} (\mathbf{y} - \mathbf{1}\hat{\mu})}{N}. \quad (5)$$

Then, θ and $\mathbf{p}$ can be estimated by maximizing (3). For test point $\mathbf{x}^*$, the best linear unbiased predictor is

$$\tilde{y}(\mathbf{x}^*) = \hat{\mu} + \mathbf{r}^T \mathbf{C}^{-1} (\mathbf{y} - \mathbf{1}\hat{\mu}) \quad (6)$$

$$s^2(\mathbf{x}^*) = \hat{\sigma}^2 \left[1 - \mathbf{r}^T \mathbf{C}^{-1} \mathbf{r} + \frac{(1 - \mathbf{1}^T \mathbf{C}^{-1} \mathbf{r})^2}{\mathbf{1}^T \mathbf{C}^{-1} \mathbf{1}} \right] \quad (7)$$

where $\mathbf{r}^T = [\text{corr}(\epsilon(\mathbf{x}^*), \epsilon(\mathbf{x}^1)), \dots, \text{corr}(\epsilon(\mathbf{x}^*), \epsilon(\mathbf{x}^N))]$, $\tilde{y}(\mathbf{x}^*)$ is the prediction mean, and $s^2(\mathbf{x}^*)$ is the prediction uncertainty.

B. LCB Prescreening Strategy

In the LCB prescreening strategy [19], [24], [25], the prediction value and uncertainty given by (6) and (7) are combined as

$$y_{\text{LCB}}(\mathbf{x}^*) = \tilde{y}(\mathbf{x}^*) - \omega s(\mathbf{x}^*), \quad \omega \in [0, 3] \quad (8)$$

where $s(\mathbf{x}^*)$ is the square root of $s^2(\mathbf{x}^*)$. A qualitative analysis is given that balances the exploration and exploitation of the LCB constant ω . More details are shown in [22].

Let us consider two extreme cases. In case I, the test point $\mathbf{x}^*$ is very far from every observed point, $\mathbf{r} \approx \mathbf{0}$, and s is approximately σ . In case II, the test point $\mathbf{x}^*$ is identical to one observed point. Let $\mathbf{x}^* = \mathbf{x}^i$ and $\mathbf{r} = \mathbf{r}(\mathbf{x}^*, \mathbf{X}) = \mathbf{r}(\mathbf{x}^i, \mathbf{X})$. $\mathbf{r}(\mathbf{x}^i, \mathbf{X})$ is equal to the i th column of the matrix $\mathbf{C}$, which can be described as

$$\mathbf{r}(\mathbf{x}^i, \mathbf{X}) = \mathbf{C} \mathbf{e}_i \quad (9)$$

where $\mathbf{e}_i$ is the i th unit vector. We multiply both sides of (9) by $\mathbf{C}^{-1}$ and obtain $\mathbf{C}^{-1} \mathbf{r} = \mathbf{e}_i$. The second term of (7) is expressed by

$$\mathbf{r}^T \mathbf{C}^{-1} \mathbf{r} = \mathbf{r}^T \mathbf{e}_i \equiv \text{corr}(\epsilon(\mathbf{x}^i), \epsilon(\mathbf{x}^i)) = 1 \quad (10)$$

and the last term is given by

$$\mathbf{1}^T \mathbf{C}^{-1} \mathbf{r} = \mathbf{1}^T \mathbf{e}_i = 1. \quad (11)$$

Then, we have $s = 0$. The size of s depends on the distance between the test point and the observed points, i.e., the correlation between them. The LCB prescreening strategy is used in minimization problems. When ω is small, the optimal solution is more likely to exploit the current promising region; when ω is large, the searching path trends toward unexplored areas, where the observed points are sparse.

C. Multifidelity GPR

The HF model yields accurate simulation results but is time-consuming, and the LF model has a short simulation time but low accuracy. To learn the relationships among different fidelities, a small amount of HF data is used to train the surrogate model to reduce the computational burden.

For multifidelity models, a recursive Gaussian process [26] can be used to establish a regression surrogate model. The fidelity training set of T levels is organized as $\mathbb{D}_t = \{\mathbf{X}_t, \mathbf{Y}_t\}$, $t = 1, \dots, T$, and the subscript from small to large represents the fidelity level from low to high. $\mathbf{X}_t$ and $\mathbf{Y}_t$ represent the input and output of the training set, respectively, and $\mathbf{Y}_t$ denotes the simulation data of the t -level fidelity model. Inspired by deep learning [27], an autoregressive multifidelity model is proposed, which can learn complex nonlinear correlations between HF and LF data and is described as

$$\begin{aligned} f_t(\mathbf{X}_t) &= g_t(\tilde{\mathbf{X}}_t), \quad 2 \leq t \leq T \\ \tilde{\mathbf{X}}_t &= [\mathbf{X}_t, f_{*t-1}(\mathbf{X}_t)] \end{aligned} \quad (12)$$

where function g_t is a Gaussian process model that maps the LF model to the HF model and the GP prior f_{t-1} is replaced by the GP posterior f_{*t-1} . The training and prediction process will be introduced in detail as follows. Without loss of generality, we suppose that there are two fidelity levels, denoted as $\mathbb{D}_L = [\mathbf{X}_L, \mathbf{Y}_L]$ and $\mathbb{D}_H = [\mathbf{X}_H, \mathbf{Y}_H]$, $\mathbf{X}_H \subset \mathbf{X}_L$, where the subscript L represents LF data and H represents HF data. For two types of fidelity data, the multifidelity GPR constructs two surrogate models. First, the LF data are utilized to construct the first surrogate model, denoted as $\mathbf{U}_L^s$. The training process and the prediction process of the first surrogate model are identical to the ordinary GPR as introduced in Section II-A. Subsequently, the second surrogate model is constructed based on $\tilde{\mathbb{D}}_H = [\tilde{\mathbf{X}}_H, \mathbf{Y}_H]$, where $\tilde{\mathbf{X}}_H = [\mathbf{X}_H, \mathbf{U}_L^s(\mathbf{X}_H)]$ and $\mathbf{U}_L^s(\mathbf{X}_H)$ represents the predicted value of $\mathbf{X}_H$ with the first surrogate model $\mathbf{U}_L^s$. The input vector of the training set is $\tilde{\mathbf{X}}_H$. The prediction process is given as follows.

- 1) Use the first surrogate model $\mathbf{U}_L^s$ to obtain the prediction value $\hat{y}_L$ and prediction uncertainty $\hat{s}_L$ of the test point $\mathbf{x}^*$.
- 2) Generate an $I \times 1$ random vector $\mathbf{v}$ with every element $v_i \sim \mathcal{N}(0, 1)$.
- 3) Predict

$$[\tilde{\mathbf{y}}_H, \tilde{\mathbf{s}}_H] = \begin{bmatrix} \mathbf{U}_H^s(\mathbf{x}^*, \hat{y}_L + v_1 \hat{s}_L) \\ \vdots \\ \mathbf{U}_H^s(\mathbf{x}^*, \hat{y}_L + v_I \hat{s}_L) \end{bmatrix} \quad (13)$$

where $[\tilde{\mathbf{y}}_H, \tilde{\mathbf{s}}_H] \in \mathbb{R}^{I \times 2}$. The final predicted value is

$$\hat{y}_H = m(\tilde{\mathbf{y}}_H) \quad (14)$$

and predicted uncertainty is

$$\hat{s}_H^2 = m(\tilde{\mathbf{s}}_H^2) + \text{var}(\tilde{\mathbf{y}}_H) \quad (15)$$

where $m(\cdot)$ is the mean and $\text{var}(\cdot)$ is the variance.

D. MB-MLAO Framework

The schematic of the proposed MB-MLAO method is shown in Fig. 2, where $\mathbf{U}_L$, $\mathbf{U}_H$, and $\mathbf{U}_s$ represent the LF EM simulation model, HF EM simulation model, and surrogate model, respectively. The flow diagram is shown in Fig. 3, and the steps of the MB-MLAO procedure are shown as follows.

Step 1 (Initialize Settings): Define the design variables, optimized specifications, search space $[a, b]^K$, the

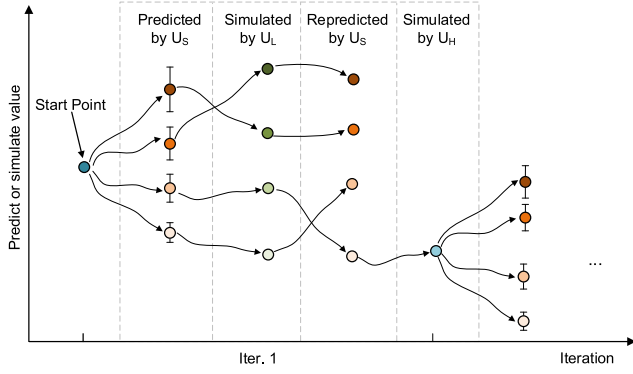


Fig. 2. Schematic of the proposed MB-MLAO method.

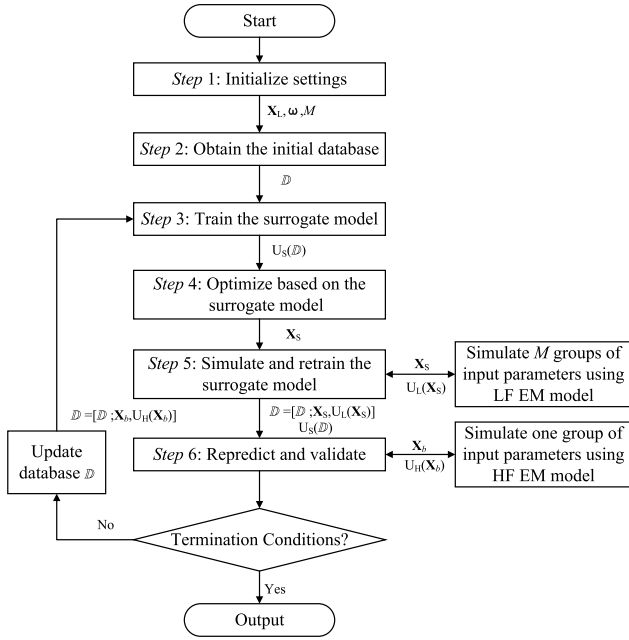


Fig. 3. Flow diagram of the proposed MB-MLAO method.

number of LCB constants M , and their values $\omega = \omega_i, i = 1, \dots, M$. N_L parameter vectors are sampled by Latin hypercube sampling (LHS) [28] within the searching space, denoted as $\mathbf{X}_L$.

Step 2 (Obtain the Initial Database): Calculate $\mathbf{X}_L$ by an LF EM simulation and obtain the response $\mathbf{R}_L = \mathbf{U}_L(\mathbf{X}_L)$. Select N_H parameter vectors from $\mathbf{X}_L$, which are denoted as $\mathbf{X}_H$, calculate $\mathbf{X}_H$ by an HF EM simulation, and obtain $\mathbf{R}_H = \mathbf{U}_H(\mathbf{X}_H)$.

Step 3 (Train the Surrogate Model): Use multifidelity GPR learning for the relationship between the input parameters and the response to obtain an inexpensive surrogate model $\mathbf{U}_S$.

Step 4 (Optimize Based on the Surrogate Model): Use a global algorithm such as a GA to optimize the surrogate model constructed in Step 3. The fitness function can be expressed as

$$F(\mathbf{x}_i) = \min_{\mathbf{x}_i} \{\tilde{y}(\mathbf{x}_i) - \omega_i \hat{s}(\mathbf{x}_i)\}, i = 1, \dots, M. \quad (16)$$

Obtain the optimal values and input parameters $\mathbf{X}_S = \{\mathbf{x}_i | i = 1, \dots, M\}$ under M groups of different branches.

Step 5 (Simulate and Retrain the Surrogate Model): To control the additional computational budget and improve the prediction accuracy, use the LF EM simulation to calculate the above M groups of input parameters $\mathbf{X}_S$. Then, add the M groups LF data into the database $\mathbb{D}$ and retrain the surrogate model.

Step 6 (Repredict and Validate): By repredicting the M group input parameters, due to the existence of LF data, the prediction accuracy of the M group input parameters is likely improved. Select the input parameter vector corresponding to the optimal repredicted value for the HF EM model to verify. Note that the validated input parameter vector is the algorithm adaptively selected from M groups of parameters. If the termination condition such as the optimized target or maximum number of iterations is satisfied, stop the loop, otherwise, update the database and repeat Step 3.

III. VERIFICATION USING TEST FUNCTIONS

Two test functions are used to illustrate the superiority of the MB-MLAO algorithm over the conventional single-branch MLAO (SB-MLAO) algorithm. The K -dimensional, multimodal and variable-fidelity test problem Ackley (Function I) [29] and K -dimensional, unimodal and variable-fidelity test problem Ellipsoid (Function II) are used here. Function I is defined by

$$\begin{cases} y_H = -20 \exp \left\{ -\frac{1}{5} \sqrt{\frac{1}{K} \sum_{i=1}^K (x_i - a_i)^2} \right\} \\ \quad - \exp \left\{ \frac{1}{K} \sum_{i=1}^K \cos(2b\pi(x_i - a_i)) \right\} + 20 + \exp\{1\} \\ y_L = -20 \exp \left\{ -\frac{1}{5} \sqrt{\frac{1}{K} \sum_{i=1}^K x_i^2} \right\} \\ \quad - \exp \left\{ \frac{1}{K} \sum_{i=1}^K \cos 2\pi x_i \right\} + 20 + \exp\{1\} \end{cases} \quad (17)$$

where $K = \{5, 10, 20\}$, $b = 1.3$, and $\mathbf{a} = [1.2, 0.2, 1.4, 0.8, 1.8, 1.0, 1.6, 0.6, 2.0, 0.4, 1.3, 0.3, 1.5, 0.9, 1.9, 1.1, 1.7, 0.7, 2.1, 0.5]$. In addition, Function II is expressed as

$$\begin{cases} y_H = \sum_{i=1}^5 i(x_i - a_i)^2 \\ y_L = \sum_{i=1}^5 i x_i^2 \end{cases} \quad (18)$$

where $K = \{5, 10\}$ and $\mathbf{a} = [0.1, 0.4, 0.5, 0.3, 0.2, 0.6, 0.6, 0.2, 0.8, 1.0]$.

Several test functions are examined in [30]. Simulation results demonstrate that a large quantity of LF data improves the accuracy of the LF surrogate model and beneficial to

TABLE I
SETTINGS OF FUNCTION I AND MEAN OF NRMSE
(MNMSE) OF TEN EXPERIMENTS

Dimension and Search Range	N_H	N_L	MNMSE	
			Case I	Case II
$[-3, 3]^5$	20	100 (101)	0.1082	0.0942
$[-3, 3]^{10}$	20	100 (101)	0.0769	0.0702
$[-3, 3]^{20}$	50	100 (101)	0.0626	0.0446

the prediction of the multifidelity model but only for highly linearly correlated functions. However, if the linear correlation of LF and HF data is poor, improvements in LF surrogate models will not help the predictions of multifidelity models and may lead to worse predictions. The multifidelity GPR in Section II-C has good learning ability for cases with nonlinear LF and HF data [27]. To compare the differences between the predicted data with and without the corresponding LF data, Function I with dimensions of 5, 10, and 20 is tested. All experiments are run ten times. Table I presents the statistical results with and without the corresponding LF data. After the surrogate model has been trained, case I indicates that the test data are directly predicted. Case II indicates that the test data are first calculated using the LF model, and the LF data are added to the training set for retraining; then, the test data are predicted. There are 200 sets of test data for each experiment. For each set of test data, case II has only one more LF data point than case I. A typical result of all experiments is shown in Fig. 4, and the blue dots and red dots represent cases I and II, respectively. The blue shade and red shade represent the area covering 95% of the predicted points for 200 sets of test data for cases I and II, respectively. The red dots are more concentrated on the black line in the middle than the blue dots. However, the prediction with LF data may be poorer than that without LF data such as the points circled by dotted lines in Fig. 4(c) and (d). As shown in Table I, after retraining, the normalized root-mean-square error (NRMSE) of ten experiments has a smaller mean value, which implies that the accuracy of reprediction is likely improved.

We assume that the calculation time of the HF model τ_{HF} , training time τ_{train} , optimization time τ_{opti} , and calculation time of the LF model τ_{LF} follows the relationships:

$$\tau_{HF} = 20\tau_{LF}, \quad \tau_{train} = 0.2\tau_{LF}, \quad \text{and} \quad \tau_{opti} = 0.5\tau_{LF}. \quad (19)$$

The total optimization time τ_{total} can be expressed as

$$\tau_{total} = \tau_{initial} + I(M\tau_{LF} + \tau_{HF} + \rho\tau_{train} + M\tau_{opti}) \quad (20)$$

where

$$\rho = \begin{cases} 1, & \text{SB - MLAO} \\ 2, & \text{MB - MLAO} \end{cases} \quad (21)$$

$\tau_{initial}$ is the time to prepare the initial database, M is the number of LCB variables, and I is the number of iterations of MLAO. For SB-MLAO, M is equal to 1, and $\tau_{total,S} = \tau_{initial} + 21.7I_S\tau_{LF}$. For MB-MLAO, $\tau_{total,M} = \tau_{initial} + (20.4 + 1.5M)I_M\tau_{LF}$. For Function I, the maximum of iteration of

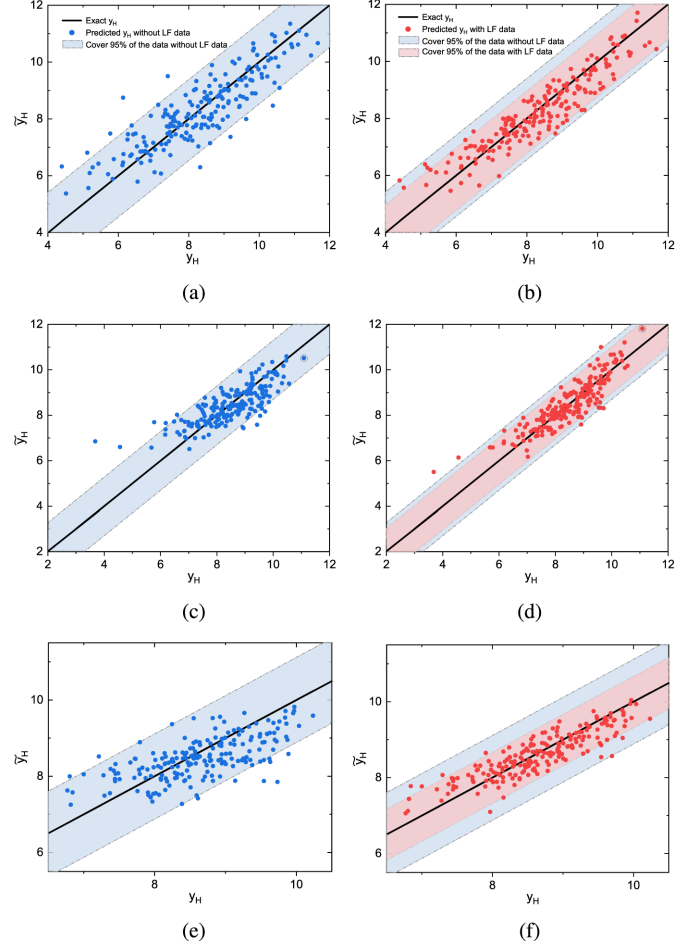


Fig. 4. Exact and predicted values of Function I. (a) $K = 5$, without LF data. (b) $K = 5$, with LF data. (c) $K = 10$, without LF data. (d) $K = 10$, with LF data. (e) $K = 20$, without LF data. (f) $K = 20$, with LF data.

SB-MLAO is set as $I_S = 100$. $\tau_{total,S} \geq \tau_{total,M}$ can be obtained if $I_{M=3} \leq 87$, $I_{M=4} \leq 82$, and $I_{M=5} \leq 77$. For Function II, the maximum number of iterations of SB-MLAO is set as $I_S = 50$. The condition $I_{M=3} \leq 43$, $I_{M=4} \leq 41$, and $I_{M=5} \leq 38$ should be satisfied such that $\tau_{total,S} \geq \tau_{total,M}$.

The results of MB-MLAO and SB-MLAO are shown in Fig. 5. For Functions I and II, ten independent runs are performed in MATLAB. Q_1 is the first quartile and Q_3 is the third quartile. From Tables II and III, MB-MLAO basically obtained better results than SB-MLAO on 10-D test functions. On 5-D test functions, SB-MLAO with $\omega = 1.5$, 2, and 3 has better results. However, MB-MLAO maintains a similar optimization efficiency with $\omega = 1$ and is better than $\omega = 0$ and 0.5. The selection of ω affects the performance of SB-MLAO. An inappropriate LCB constant may cause the algorithm to slowly converge or fall into a local optimum, and the robustness of the algorithm is reduced. Thus, for problems with different features and complexities, MB-MLAO maintains effectiveness while obtaining robustness.

IV. APPLICATION TO ANTENNA DESIGN

Three antennas are used to verify the superiority of the proposed algorithm. The function libraries of Emukit [33] and Geatpy [34] based on Python are used here.

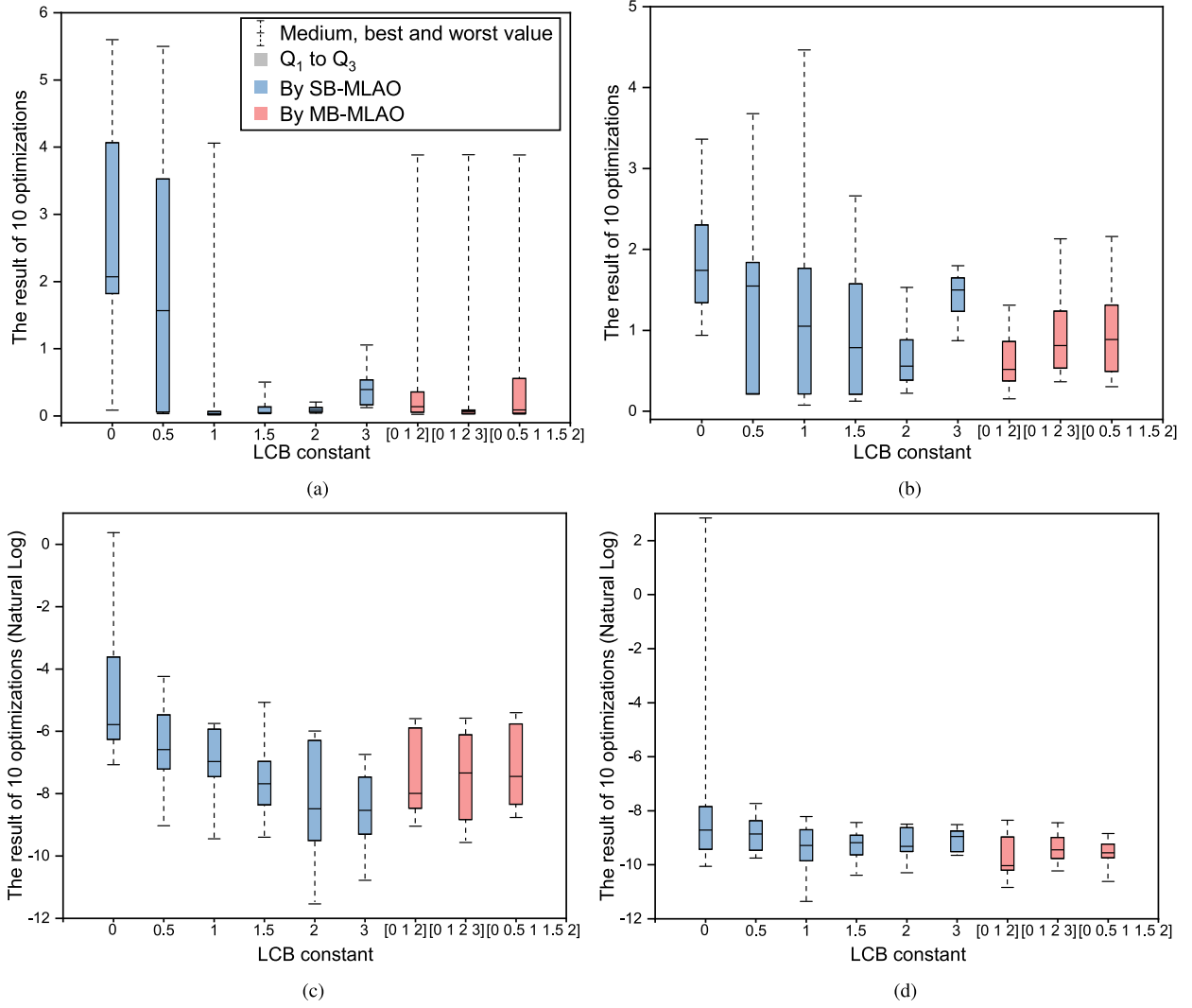


Fig. 5. Medium, best, and worst value of the test functions in ten optimizations. (a) Function I with $[-3, 3]^5$. (b) Function I with $[-3, 3]^{10}$. (c) Function II with $[-3, 3]^5$. (d) Function II with $[-3, 3]^{10}$.

TABLE II
AVERAGE AND STANDARD DEVIATION (STD.) OF
TEN OPTIMIZATIONS OF FUNCTION I

ω	$[-3, 3]^5$		$[-3, 3]^{10}$	
	Mean	Std.	Mean	Std.
0	2.75E+00	1.77E+00	1.87E+00	7.08E-01
0.5	2.08E+00	1.90E+00	1.46E+00	1.12E+00
1	4.71E-01	1.27E+00	1.38E+00	1.36E+00
1.5	1.30E-01	1.56E-01	1.01E+00	9.15E-01
2	9.79E-02	5.02E-02	6.67E-01	4.07E-01
3	4.13E-01	2.88E-01	1.46E+00	2.79E-01
[0, 1, 2]	6.21E-01	1.21E+00	6.23E-01	3.51E-01
[0, 1, 2, 3]	4.39E-01	1.21E+00	9.75E-01	5.99E-01
[0, 0.5, 1, 1.5, 2]	5.78E-01	1.18E+00	1.01E+00	6.25E-01

TABLE III
AVERAGE AND STANDARD DEVIATION (STD.) OF TEN
OPTIMIZATIONS OF FUNCTION II

ω	$[-3, 3]^5$		$[-3, 3]^{10}$	
	Mean	Std.	Mean	Std.
0	2.95E-01	6.09E-01	1.82E+00	5.37E+00
0.5	3.09E-03	4.25E-03	1.69E-04	1.16E-04
1	1.38E-03	1.15E-03	1.08E-04	7.79E-05
1.5	1.15E-03	1.89E-03	1.05E-04	5.60E-05
2	7.71E-04	1.02E-03	1.13E-04	6.02E-05
3	3.91E-04	4.24E-04	1.26E-04	4.96E-05
[0, 1, 2]	1.23E-03	1.48E-03	7.96E-05	7.15E-05
[0, 1, 2, 3]	1.19E-03	1.39E-03	9.72E-05	5.55E-05
[0, 0.5, 1, 1.5, 2]	1.66E-03	1.73E-03	7.84E-05	3.46E-05

A. Ultrawideband Planar Dipole Antenna (Antenna 1)

The ultrawideband (UWB) antenna proposed in [31] is fabricated with a Rogers 5880 substrate of relative permittivity

$\epsilon_r = 2.2$ and with $h = 1.58$ mm. Its geometry is shown in Fig. 6. The design variables are $[l_0, w_0, a_0, l_p, w_p, s_0]$. The low bounds are $[18, 12, 0.3, 12, 5, 0.8]$ mm, and the high bounds are $[20, 14, 0.7, 14, 7, 1.2]$ mm.

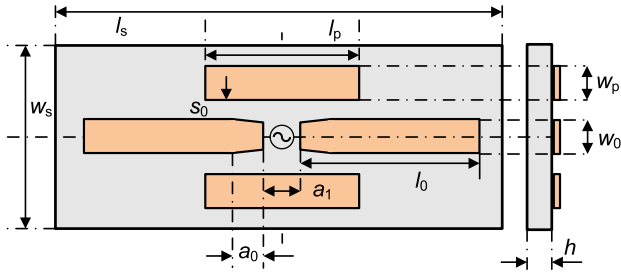


Fig. 6. Geometry of the UWB antenna (Antenna 1) [31].

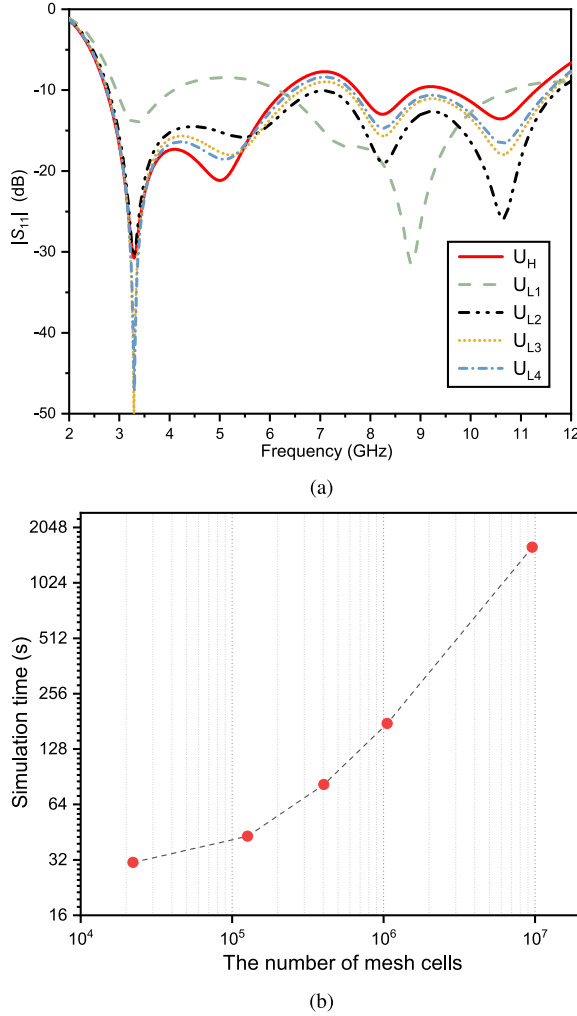


Fig. 7. Antenna 1 simulated through CST Microwave Studio [32]. (a) Reflection coefficients $|S_{11}|$ with different mesh cells. U_H : 9596580 mesh cells and 1600 s simulation time. U_{L1} : 22176 mesh cells and 31 s simulation time. U_{L2} : 126360 mesh cells and 43 s simulation time. U_{L3} : 404040 mesh cells and 82 s simulation time. U_{L4} : 1058400 mesh cells and 176 s simulation time. (b) Simulation time against the number of mesh cells.

For UWB antennas, LF models are generally obtained through coarse-mesh EM models. The LF model must capture important features in the HF model [15]. Different LF EM simulation models and different initial sample sets are compared to illustrate the superiority of the proposed method. Fig. 7 shows the curves of $|S_{11}|$ of different mesh cells and their simulation time. U_H represents the HF model. U_{L1} is too

coarse to capture the feature of U_H . U_{L3} and U_{L4} are properly representing $|S_{11}|$ of U_H . U_{L3} and U_{L4} have similar response curves, but U_{L3} is twice as fast as U_{L4} . U_{L3} is a reasonable choice to replace U_{L4} as an LF model. Although U_{L2} is not as accurate as U_{L3} , its simulation time is shorter. Next, U_{L2} and U_{L3} are used in optimization. The optimization goal is to maximize $|S_{11}|$ to less than -10 dB within 3–11 GHz.

The LF training data input parameter vector $\mathbf{X}_L$ is sampled by the LHS method with $N_L = 40$, and HF training data are selected from $\mathbf{X}_L$ with $N_H = 5$. The maximum within bandwidths of $|S_{11,i}^L|$, $i = 1, \dots, N_L$ and $|S_{11,i}^H|$, $i = 1, \dots, N_H$ are used as the output of the surrogate model. The fitness function can be described as

$$f(\mathbf{x}_i) = \min_{\mathbf{x}_i} \{ \tilde{R}_S(\mathbf{x}_i) - \omega_i R_S^s(\mathbf{x}_i) \}, \quad i = 1, 2, \dots, M \quad (22)$$

where $\tilde{R}_S(\mathbf{x}_i)$ and $R_S^s(\mathbf{x}_i)$ represent the predicted value and predicted uncertainty, respectively. The termination condition is reaching the optimization goal or 50 iterations. Fig. 8(c), (f), and (i) shows the changes in LCB variables corresponding to the HF EM simulation at each iteration of MB-MLAO. In the MB-MLAO method, multiple LCB variables are denoted as $\omega = [0, 1, 2]$ and compared with the LCB constant in the SB-MLAO method, where ω is equal to 0, 1, or 2, respectively. Fig. 8 shows the convergence curves of different initial sample sets, different LM models, and their $|S_{11}|$. The LF EM model of Fig. 8(a) and (d) is U_{L2} . The initial input vectors are denoted as $\mathbf{X}_L^1$ and $\mathbf{X}_H^1$ and $\mathbf{X}_L^2$ and $\mathbf{X}_H^2$. U_{L3} is used as an LF EM model in Fig. 8(g). In Fig. 8(a), the optimal LCB constant is $\omega = 2$. However, in Fig. 8(d) and (g), the case of $\omega = 2$ is the worst. In this example, the proposed MB-MLAO method can achieve the goal within 9 h with different initial sample sets and LF models, and MB-MLAO is more robust than SB-MLAO.

B. Triband Patch Antenna (Antenna 2)

A triband antenna is utilized to verify the MB-MLAO method. The geometry of the triantenna is shown in Fig. 9, and more design details are provided in [35]. The design variables are $[l_2, l_{s1}, l_{s2}, l_{s3}, w_{s1}, w_{s2}, l_p, w_p, l_{stub}]$. The low bounds are [5, 19, 20, 5, 0.2, 1.5, 32, 31, 16] mm, and the high bounds are [9, 26, 25, 7.5, 2.2, 2.5, 35, 34, 20] mm.

The HF simulation model has approximately 2 million mesh cells, and the computation time is 960 s. The LF simulation model uses approximately 109 thousand mesh cells with a computational time of 94 s. Curves of $|S_{11}|$ of LF and HF EM models are shown in Fig. 10. The LF training data input parameter vector is sampled by the LHS method with $N_L = 50$. The selection of N_H will influence the sampling computation burden and accuracy of the surrogate model. The ratio between HF and LF datasets is defined as $R_{H,L} = (N_H/N_L) \times 100\%$. The optimization procedures are performed for MB-MLAO and SB-MLAO with $R_{H,L} \in \{10\%, 20\%\}$, i.e., $N_H \in \{5, 10\}$.

The optimization goal is for the $|S_{11}|$ values of 2.5, 3.5, and 5.2 GHz to be less than -10 dB. A surrogate model, U_j ($j = 1, 2, 3$), is trained for each frequency point, and the output of each surrogate model is $|S_{11}|$ of corresponding frequency point. The optimization problem can be converted such that the

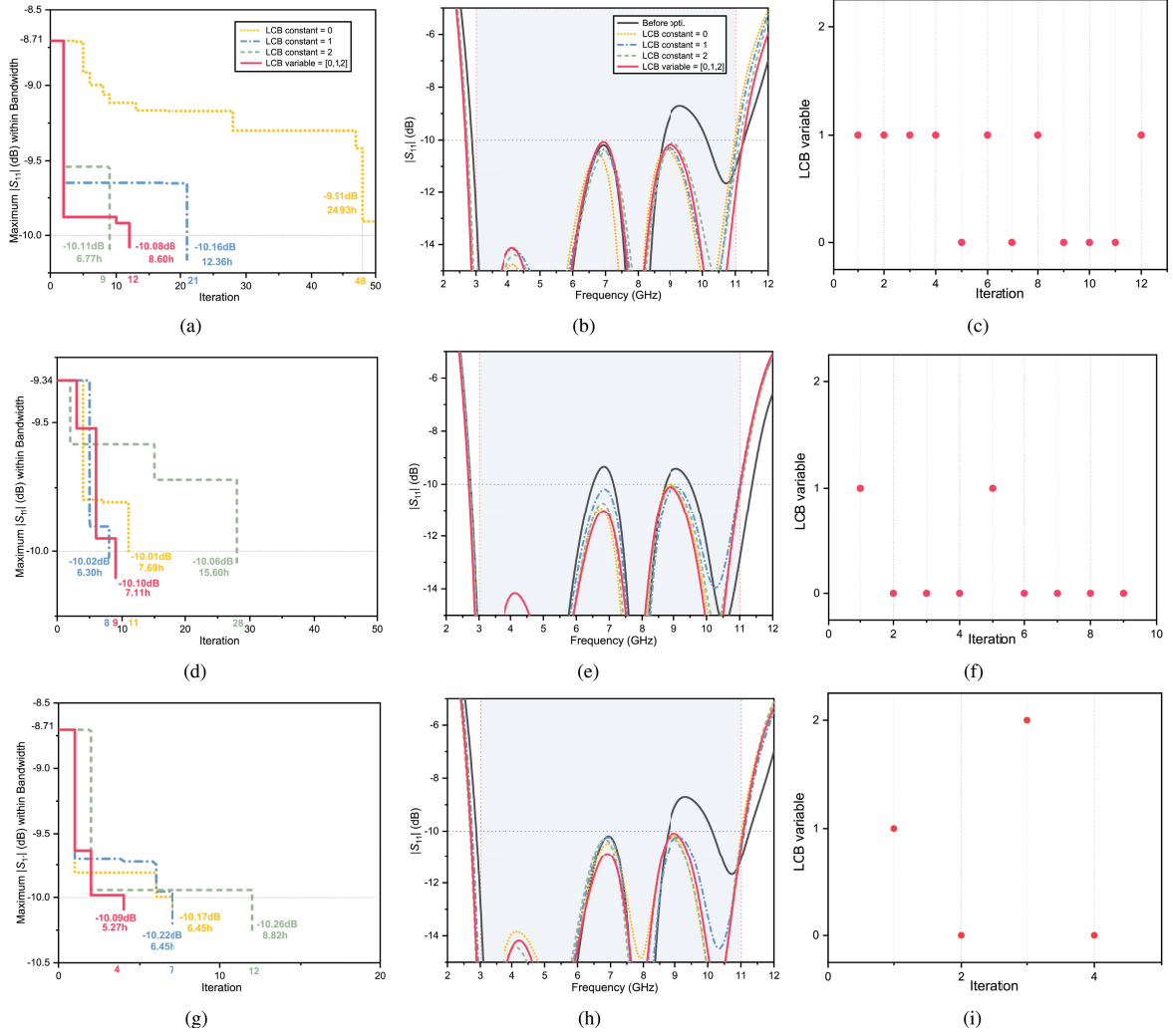


Fig. 8. Convergence curves and the reflection coefficients for Antenna 1 using different methods. Changes in LCB variables corresponding to the HF EM simulation at each iteration of MB-MLAO. (a)–(c) $\mathbf{X}_L^1$, $\mathbf{X}_H^1$, and $\mathbf{U}_{L2}$, respectively. (d)–(f) $\mathbf{X}_L^2$, $\mathbf{X}_H^2$, and $\mathbf{U}_{L2}$, respectively. (g)–(i) $\mathbf{X}_L^1$, $\mathbf{X}_H^1$, and $\mathbf{U}_{L3}$, respectively.

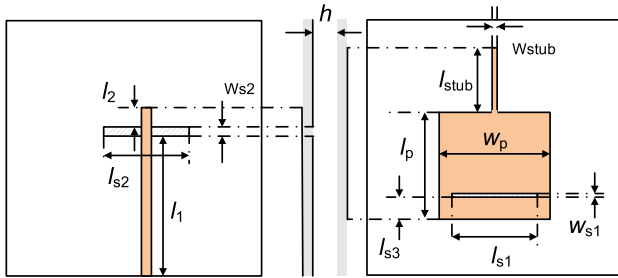


Fig. 9. Geometry of the triband patch antenna (Antenna 2) [35].

maximum of the three frequency points is less than -10 dB as follows:

$$f(\mathbf{x}_i) = \min_{\mathbf{x}_i} \left\{ \max_j \left\{ \tilde{R}_{S,j}(\mathbf{x}_i) - \omega_i R_{S,j}^s(\mathbf{x}_i) \right\} \right\} \quad (23)$$

where $i = 1, \dots, M$. $\tilde{R}_{S,j}(\mathbf{x}_i)$ is the predicted value of $\mathbf{x}_i$ by $\mathbf{U}_j$. Fig. 11 shows some typical results. As shown in Fig. 11(a), only $\omega = 0$ and $\omega = [0, 1, 2]$ reach the optimization

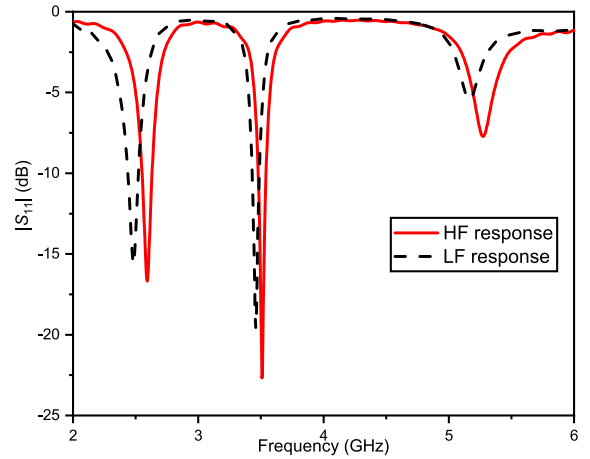


Fig. 10. HF and LF response curves of the reflection coefficients of Antenna 2.

target within the specified number of 100 iterations. When $N_L = 50$ and $N_H = 10$, $\omega = 0$, $\omega = 1$, and $\omega = [0, 1, 2]$ reach

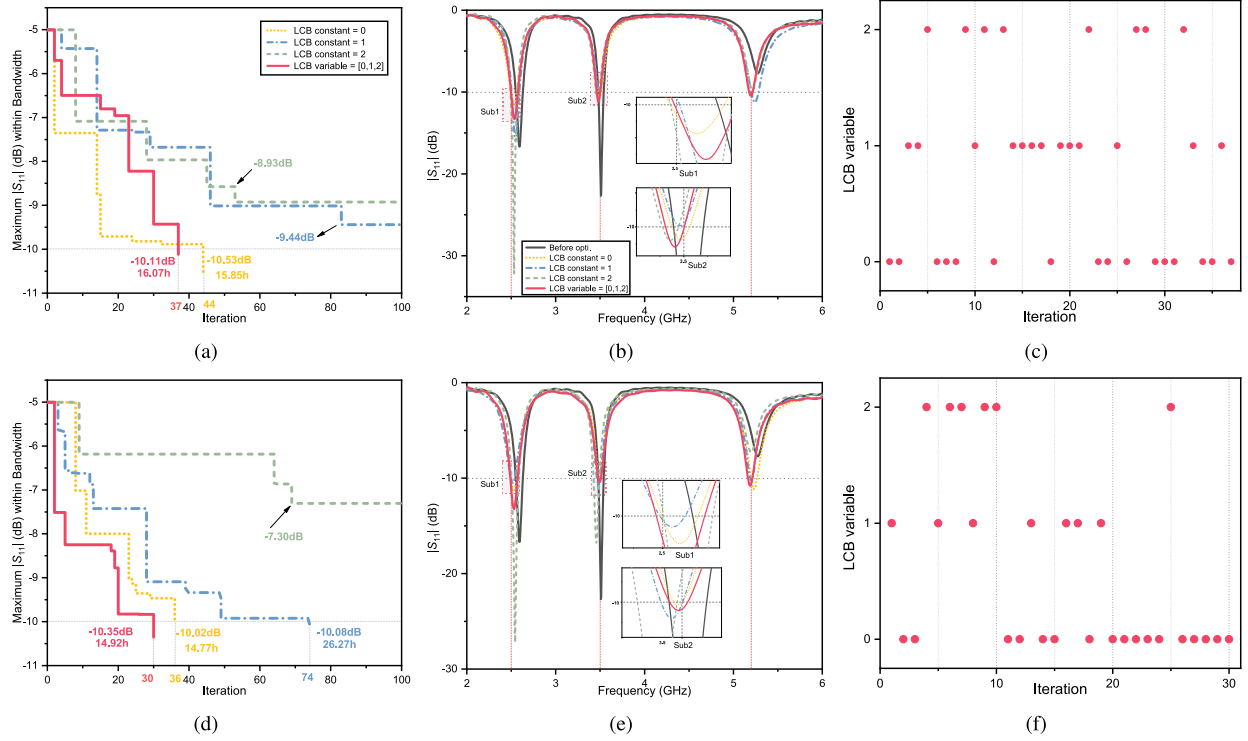


Fig. 11. Convergence curves and the reflection coefficients for Antenna 2 using different methods. Changes in LCB variables corresponding to the HF EM simulation at each iteration of MB-MLAO. (a)–(c) $N_L = 50$ and $N_H = 5$. (d)–(f) $N_L = 50$ and $N_H = 10$.

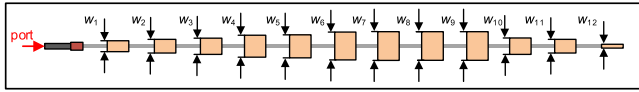


Fig. 12. Geometry of a series-fed microstrip array antenna (Antenna 3).

the optimization target. In the two experiments, MB-MLAO shows a comparable convergence speed to the SB-MLAO with $\omega = 0$. However, when the optimal ω is prior unknown, the selection of ω affects the optimization results. For example, $\omega = 2$ does not reach the target in either test. The proposed method is more robust than the SB-MLAO method when the ratio of HF and LF data varies for the same antenna.

C. Series-Fed Microstrip Array Antenna (Antenna 3)

Antenna 3 is fabricated on a single-layer Rogers RO3003 and designed for 77 GHz automotive radar applications with 12 series-fed microstrip antenna elements [36]. Its worst case performance (WCP) is determined using the proposed MB-MLAO. In the 77 GHz band, the size of the antenna is smaller, and the manufacturing tolerance requirement is stricter. WCP searching is a popular method to analyze the sensitivity of antennas when manufacturing tolerances are known prior [2]. The geometry of Antenna 3 is shown in Fig. 12. The design variables are $[w_1, w_2, w_3, w_4, w_5, w_6, w_7, w_8, w_9, w_{10}, w_{11}, w_{12}]$. The design points are $\mathbf{x}_0 = [0.56, 0.7, 0.84, 1.12, 1.19, 1.4, 1.47, 1.47, 1.47, 0.84, 0.7, 0.21]^T$ mm. The input tolerances are $\delta = [0.1, 0.1, 0.1, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.1, 0.1, 0.05]^T$ mm. The searching range is $\mathbf{x}_0 \pm \delta$.

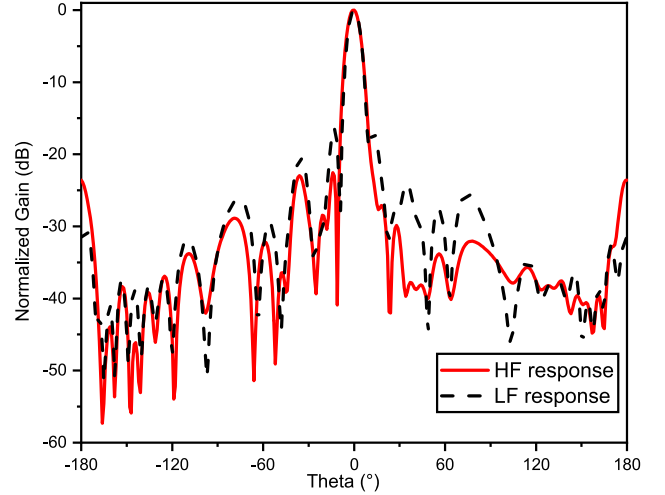


Fig. 13. HF and LF response curves of the radiation patterns of Antenna 3.

The HF responses are simulated in CST Microwave Studio with approximately 5 million mesh cells, and the computation time is 0.52 h. The LF responses are simulated with approximately 201 thousand mesh cells, and the computation time is reduced to 163 s. Curves of radiation pattern of LF and HF EM models are shown in Fig. 13. The optimization target of Antenna 3 is to maximize the sidelobe level (SLL) within the input tolerance, which can be converted to a minimization problem, i.e.,

$$f(\mathbf{x}_i) = \min_{\mathbf{x}_i} \{ -\{ \tilde{R}_S(\mathbf{x}_i) + \omega_i R_S^s(\mathbf{x}_i) \} \}, \quad i = 1, 2, \dots, M. \quad (24)$$

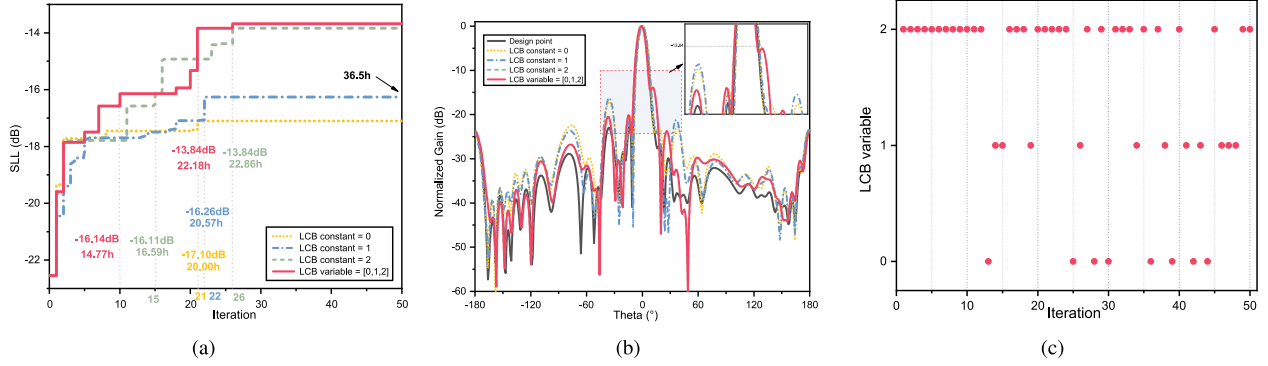


Fig. 14. Simulation results of Antenna 3 using different methods. (a) WCP searching convergence curves. (b) Normalized patterns. (c) Changes in LCB variables corresponding to the HF EM simulation at each iteration of MB-MLAO.

First, the LF input parameters are randomly sampled by the LHS method around the design point, $N_L = 50$, and HF input data are selected from LF parameter vectors, $N_H = 10$. Then, the LF and HF data of the design point are added to the initial database. Fig. 14(c) shows the changes in LCB variables of the corresponding HF EM simulation in the WCP search process. The results of different search methods are shown in Fig. 14. The search results of $\omega = [0, 1, 2]$ and $\omega = 2$ are similar, and the worst case searched around 23 h is -13.84 dB. In this experiment, in the case of $\omega = 0$ with the worst search ability, within the specified 50 iterations, the worst value searched is -17.10 dB. The value searched for $\omega = 1$ at 20.57 h is -16.26 dB, but the worst value of MB-MLAO at 14.77 h is close to this value, which is -16.14 dB. For conventional SB-MLAO method, the selection of ω will affect the search ability. As in this example, the search ability of LCB constant being equal to 2 is better than those of the LCB constant being equal to 0 and 1. The optimization results of MB-MLAO are close to the best case of SB-MLAO, and MB-MLAO is more robust.

MB-MLAO has shown robustness compared with SB-MLAO, but when the number of parameters is several dozens, the curse of dimensionality will be a main limiting factor [12]. On one hand, with the increase in dimension, the number of training samples will greatly increase. It is difficult to obtain an amount of expensive HF EM data for complex antenna problems. On the other hand, with the increase in number of samples, the training and update time of the GPR model cannot be ignored; otherwise, MB-MLAO is based on the linear combination of the prediction and predicted uncertainty of GPR. When GPR models come to high-dimensional problems, but only a small of training samples, the prediction uncertainty will be less reliable [37]. Some strategies can be performed to mitigate these problems, such as dimension reduction [18], training surrogate model cost reduction [25], and ensemble learning [38].

V. CONCLUSION

An MB-MLAO method that balances exploitation and exploration is proposed. The LCB variable is utilized to realize multibranch optimization, considering the current promising region and sampling sparse areas. A variable-fidelity model

reduces the computation time. The retraining and reprediction method is likely to improve the prediction accuracy and adaptively selects one parameter vector for verification by the HF EM model. To show the superiority of the MB-MLAO method, two test problems and three practical antennas are verified. For different antennas, the optimal LCB constant is usually *a priori* unknown. Even for the same antenna, due to different initial sample sets or different numbers of HF and LF datasets, the setting of the LCB constant affects the performance of the algorithm. For problems with different characteristics, different complexities, and different design specifications, the MB-MLAO method maintains effectiveness while obtaining robustness.

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